taken place. So do measurements, as embodied by stage-two wavefunction collapse, establish an asymmetry between past and future, between before and after a measurement is made?

These questions have stubbornly resisted complete solution and they remain controversial. Yet, through the decades, the predictive power of quantum theory has hardly been compromised. The stage one / stage two formulation of quantum theory, even though stage two has remained mysterious, predicts probabilities for measuring one outcome or another. And these predictions have been confirmed by repeating a given experiment over and over again and examining the frequency with which one or another outcome is found. The fantastic experimental success of this approach has far outweighed the discomfort of not having a precise articulation of what actually happens in stage two.

But the discomfort has always been there. And it is not simply that some details of wavefunction collapse have not quite been worked out. The *quantum measurement problem*, as it is called, is an issue that speaks to the limits and the universality of quantum mechanics. It's simple to see this. The stage one / stage two approach introduces a split between what's being observed (an electron, or a proton, or an atom, for example) and the experimenter who does the observing. Before the experimenter gets into the picture, wavefunctions happily and gently evolve according to Schrödinger's equation. But then, when the experimenter meddles with things to perform a measurement, the rules of the game suddenly change. Schrödinger's equation is cast aside and stage-two collapse takes over. Yet, since there is no difference between the atoms, protons, and electrons that make up the experimenter and the equipment he or she uses, and the atoms, protons, and electrons that he or she studies, why in the world is there a split in how quantum mechanics treats them? If quantum mechanics is a universal theory that applies without limitations to *everything*, the observed and the observer should be treated in exactly the same way.

Niels Bohr disagreed. He claimed that experimenters and their equipment *are* different from elementary particles. Even though they are made from the same particles, they are "big" collections of elementary particles and hence governed by the laws of classical physics. Somewhere between the tiny world of individual atoms and subatomic